

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12408289
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Lu; Wen-Hu et al.

Operation and maintenance device and cabinet having the same

Abstract

An operation and maintenance device includes a floating tray, an unlocking assembly, and a hook assembly. The unlocking assembly movably disposed on the floating tray, one end of the unlocking assembly includes a first mounting frame. The hook assembly connected to the unlocking assembly, the hook assembly comprising a second mounting frame and a hook connected to the second mounting frame, the second mounting frame is rotatably connected to the first mounting frame to drive the hook to rotate and be engaged with the slidable locking mechanism, the unlocking assembly is used to drive the hook to translate. In addition, the present disclosure also provides a cabinet.

Inventors: Lu; Wen-Hu (Tianjin, CN), Yin; Li-Yi (Tianjin, CN), Wang; Shu-Tong (Tianjin, CN)

Applicant: Fulian Precision Electronics (Tianjin) Co., LTD. (Tianjin, CN)

Family ID: 92534381

Assignee: Fulian Precision Electronics (Tianjin) Co., LTD. (Tianjin, CN)

Appl. No.: 18/213856

Filed: June 25, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240295139 A1	Sep. 05, 2024

Foreign Application Priority Data

CN	202310196537.9	Mar. 02, 2023
----	----------------	---------------

Publication Classification

Int. Cl.: G06F1/18 (20060101); H05K5/02 (20060101); H05K7/14 (20060101)

U.S. Cl.:

CPC H05K7/1489 (20130101); G06F1/187 (20130101); H05K5/0221 (20130101);
H05K5/0217 (20130101); H05K7/14 (20130101); H05K7/1487 (20130101);
H05K7/1488 (20130101)

Field of Classification Search

CPC: H05K (7/14); H05K (7/1489); H05K (5/0221); H05K (7/1487); H05K (5/0217); G06F (1/187)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10015892	12/2017	Chien	N/A	H05K 5/15
2011/0049317	12/2010	Peng	248/231.9	G06F 1/187
2022/0192046	12/2021	Wu	N/A	G06F 1/188

Primary Examiner: Wright; Kimberley S

Attorney, Agent or Firm: ScienBiziP, P.C.

Background/Summary

FIELD

(1) The subject matter herein generally relates to an operation and maintenance device, particularly to an operation and maintenance device for servers and a cabinet having the operation and maintenance device.

BACKGROUND

- (2) Hard disks need to be installed in a server cabinet. For rearrangement and maintenance purposes, the hard disks may be frequently installed into and removed from the cabinet.
- (3) However, when removing the hard disks from the cabinet, a top cover of the cabinet needs to be opened, or other components of the cabinet need to be removed to expose the hard disks, resulting in a low efficiency.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Implementations of the present technology will now be described, by way of example only, with reference to the attached figures, wherein:
- (2) FIG. 1 is a diagrammatic view of a cabinet according to an embodiment of the present disclosure.
- (3) FIG. 2 is an exploded view of the cabinet of FIG. 1.
- (4) FIG. 3 is a diagrammatic view of a storage tray of the cabinet of FIG. 1.

- (5) FIG. 4 is a diagrammatic view of an operation and maintenance device of the cabinet of FIG. 1.
- (6) FIG. 5 is a diagrammatic view of a base of the operation and maintenance device of FIG. 4.
- (7) FIG. 6 is a diagrammatic view of a floating tray of the operation and maintenance device of FIG. 4.
- (8) FIG. 7 is a diagrammatic view of an unlocking assembly of the operation and maintenance device of FIG. 4.
- (9) FIG. 8 is a diagrammatic view of a hook assembly of the operation and maintenance device of FIG. 4.
- (10) FIG. 9 is a diagrammatic view of the operation and maintenance device of FIG. 4 in a first state.
- (11) FIG. 10 is a diagrammatic view of the operation and maintenance device of FIG. 4 in a second state.
- (12) FIG. 11 is a cross-sectional view along line XI-XI of FIG. 10.

DETAILED DESCRIPTION

(13) It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different FIG.s to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale and the proportions of certain parts have been exaggerated to better illustrate details and features of the present disclosure.

(14) Referring to FIGS. 1 and 2, an embodiment of the present application provides a cabinet **100**, which may be used to install hard disks (not labeled) therein.

(15) The cabinet **100** includes a chassis **10**, a storage tray **20**, a slidable locking mechanism **30**, and an operation and maintenance device **40**. The storage tray **20** can be used to install the hard disks. The storage tray **20** is detachably fixed in the chassis **10** through the slidable locking mechanism **30**. The device **40** can be pushed into the chassis **10**, unlock the slidable locking mechanism **30**, and pull out the storage tray **20** from the chassis **10**.

(16) Referring to FIGS. 2 and 8, the device **40** includes a base **41**, a floating tray **42** movably disposed on the base **41**, an unlocking assembly **43** movably disposed on the floating tray **42**, and a hook assembly **44** disposed at one end of the unlocking assembly **43**. The unlocking assembly **43** includes a first mounting frame **431**. The hook assembly **44** includes a second mounting frame **441** and a hook **442** connecting the second mounting frame **441**. The hook **442** can be engaged with or disengaged from the slidable locking mechanism **30**. The second mounting frame **441** is rotatably connected to the first mounting frame **431**, so that the unlocking assembly **43** can rotate the hook **442** during movement, thereby switching the hook **442** between a first state A and a second state B.

(17) Referring to FIG. 9, the first state A is that the hook **442** is disengaged from the slidable locking mechanism **30**, and the storage tray **20** is connected to the chassis **10** through the slidable locking mechanism **30**. Referring to FIG. 10, the second state B is that the hook **442** is engaged with the slidable locking mechanism **30**, the slidable locking mechanism **30** is disconnected from the storage tray **20** and the chassis **10**, and the storage tray **20** may be separated from the chassis **10** by pulling the unlocking assembly **43** outward.

(18) The cabinet **100** provided by the present application includes the device **40** that can be pushed into the chassis **10**. The device **40** is provided with the hook **442** that can be driven by the unlocking assembly **43**. The hook **442** can be engaged with the slidable locking mechanism **30** to unlock the storage tray **20** and the chassis **10**. Thereby, the storage tray **20** can be separated from the chassis **10** by pulling the unlocking assembly **43** outward, which is conducive to the installation

of the storage tray **20** into the chassis **10** and the removal of the storage tray **20** from the chassis **10**, and thereby improving the operation and maintenance efficiency of the hard disks.

(19) Referring to FIGS. **2** and **3**, in this embodiment, the slidable locking mechanism **30** includes an unlocking slider **31** set in the storage tray **20** and an elastic pin **32** that goes through the unlocking slider **31**. The unlocking slider **31** defines a slanted slot **311** for the elastic pin **32** to go through and a protrusion **312** for engaging with the hook **442**. An extending direction of an inner wall of the slanted slot **311** is inclined with an extending direction of the elastic pin **32**.

(20) The elastic pin **32** includes a first end **321** and a second end **322**. A cross-sectional dimension of the first end **321** is greater than a cross-sectional dimension of the second end **322**. A cross-sectional dimension of the slanted slot **311** is greater than the cross-sectional dimension of the second end **322** and less than the cross-sectional dimension of the first end **321**, allowing the first end **321** to abut against the slanted slot **311**, and further allowing the second end **322** to pass through the slanted slot **311** and the storage tray **20** and then to be inserted into the chassis **10**. A direction from the protrusion **312** to the slanted slot **311** is substantially parallel to an extension direction of the unlocking assembly **43**.

(21) When the slidable locking mechanism **30** locks the storage tray **20**, the second end **322** of the elastic pin **32** extends through the slanted slot **311** and the storage tray **20**, and is then inserted into the chassis **10**, thereby locking the storage tray **20** in the chassis **10**.

(22) When the slidable locking mechanism **30** unlocks the storage tray **20**, the hook **442** is engaged with the protrusion **312** and pulls the unlocking slider **31** to move, and the first end **321** of the elastic pin **32** abuts against the slanted slot **311**. As the extending direction of the slanted slot **311** is inclined with the extending direction of the elastic pin **32**, when the hook **442** pulls the unlocking slider **31** to move, the second end **322** of the elastic pin **32** moves backward to the chassis **10** and exits the chassis **10** and the storage tray **20**, thereby unlocking the storage tray **20** and the chassis **10**.

(23) Referring to FIGS. **2**, **4** and **5**, in this embodiment, the base **41** is substantially square and movably disposed within the chassis **10**. The base **41** includes a first bottom plate **411**, two first side plates **412**, a second side plate **413**, and a plurality of positioning pins **414**. The two first side plates **412** are connected to opposite sides of the first bottom plate **411**. Two ends of the second side plate **413** are connected to ends of the two first side plates **412**. The second side plate **413** is also connected to the first bottom plate **411**. The two first side plates **412**, the second side plate **413**, and the first bottom plate **411** together form a first accommodating space **S1**. The floating tray **42** is movably accommodated in the first accommodating space **S1**. The positioning pins **414** are respectively disposed at other ends of the two first side plates **412** away from the second side plate **413**. One end of the storage tray **20** facing the device **40** defines a plurality of positioning holes **201**. The positioning pins **414** are used to insert into the positioning holes **201** of the storage tray **20**.

(24) Referring to FIGS. **2**, **4** and **5**, in this embodiment, the first side plate **412** is provided with a slider **412a**, and a slider slot **101** is defined in the chassis **10**. The slider **412a** is engaged with the slider slot **101**, allowing the base **41** to movably disposed with the chassis **10** along an extension direction of the slider slot **101**. In other embodiments, the movable connection between the base **41** and the chassis **10** may also be realized by means of wheels or sliding bearings.

(25) Referring to FIGS. **2**, **4** and **5**, in this embodiment, the floating tray **42** includes a second bottom plate **421**, two first side panels **422**, a second side panel **423**, and a first handle **424**. The two first side panels **422** are connected to opposite sides of the second bottom plate **421**. Two ends of the second side panel **423** are connected to ends of the two first side panels **422**. The second side panel **423** is also connected to the second bottom plate **421**. The second bottom plate **421**, the two second side panels **423**, and the second side panel **423** together form a second accommodating space **S2**. The unlocking assembly **43** is movably disposed in the second accommodating space **S2**. The first handle **424** is disposed on a side surface of the second side panel **423** away from the

second accommodating space S2. The second side plate **413** defines an opening **413a**. The first handle **424** can pass through the opening **413a**, so as to facilitate an operator to grasp the floating tray **42** and push the floating tray **42** to move relative to the base **41**.

(26) Referring to FIGS. 5 and 6, in this embodiment, a side of the first side panel **422** opposite to the second accommodating space S2 is provided with a support column **422a**. A side of the first side plate **412** facing the first accommodating space S1 is provided with a support bar **412b**. The support column **422a** can be placed on and slide relative to the support bar **412b**, allowing the floating tray **42** and the base **41** to movably connect to each other.

(27) Referring to FIGS. 4 and 5, in this embodiment, the base **41** further includes a plurality of first fixing pins **415** and a plurality of second fixing pins **416**. The first fixing pins **415** and the second fixing pins **416** are disposed on one side of the first bottom plate **411** facing the first accommodating space S1. The first fixing pins **415** and the second fixing pins **416** are symmetrically distributed on opposite sides of the unlocking assembly **43**. The first fixing pins **415** and the second fixing pins **416** are substantially parallel to each other. That is, the first fixing pins **415** and the second fixing pins **416** are substantially distributed along a straight line.

(28) The floating tray **42** further includes a first elastic member **425** and a third fixing pin **426**. The third fixing pin **426** is disposed on one side of the second bottom plate **421** facing the second accommodating space S2. The second bottom plate **421** defines a first sliding slot **421a** and a second sliding slot **421b**, and each of the first sliding slot **421a** and the second sliding slot **421b** penetrates the second bottom plate **421**. The first fixing pin **415** movably extends through the first sliding slot **421a**, and the second fixing pin **416** movably extends through the second sliding slot **421b**. The first elastic member **425** is connected between the first fixing pin **415** and the third fixing pin **426**.

(29) The second fixing pin **416** and the second sliding slot **421b** may be used to limit a moving direction and a moving range of the floating tray **42** relative to the base **41**. The connection of the first elastic member **425** between the first fixing pin **415** and the third fixing pin **426** allows the floating tray **42** to store an elastic energy during the movement of floating tray **42** relative to the base **41**. The elastic energy can provide a buffer to a collision between the second fixing pin **416** and the sliding slot **421b**.

(30) Referring to FIGS. 4 and 6, in this embodiment, the floating tray **42** further includes a plurality of first fixing columns **427** and a plurality of second fixing columns **428**. The first fixing columns **427** and the second fixing columns **428** are disposed on one side of the second bottom plate **421** facing the second accommodating space S2. The first fixing columns **427** and the second fixing columns **428** are spaced from each other. The first fixing columns **427** are disposed at one end of the second bottom plate **421** away from the second side panel **423**. The second fixing columns **428** are spaced apart from each other on the second bottom plate **421**, along a length direction of the second bottom plate **421**. The first fixing columns **427** and the second fixing columns **428** are substantially arranged in a straight line.

(31) Referring to FIGS. 6 and 7, the unlocking assembly **43** includes a pull rod **430**, a second elastic member **433**, and a third fixing column **434**. The pull rod **430** includes a pull rod body **432**. The third fixing column **434** is connected to one side of the pull rod body **432** facing away from the second accommodating space S2. The pull rod body **432** defines a first limiting slot **432a** and a plurality of second limiting slots **432b**. The first limiting slot **432a** is disposed at one end of the pull rod body **432** close to the first mounting frame **431**. The second limiting slots **432b** are spaced apart along a length direction of the pull rod body **432**.

(32) The first fixing column **427** movably goes through the first limiting slot **432a**, and the second fixing column **428** movably goes through each of the second limiting slots **432b**. The second elastic member **433** is connected between the first fixing column **427** and the third fixing column **434**. The second fixing column **428** and the second limiting slot **432b** can be used to limit a moving direction and a moving range of the unlocking assembly **43** relative to the floating tray **42**. Two ends of the

second elastic member **433** are respectively connected to the first fixing column **427** and the third fixing column **434**, so that when the unlocking assembly **43** moves relative to the base **41**, an elastic energy can be stored, and the collision between the second fixing column **428** and the second limiting slot **432b** can be buffered.

(33) Referring to FIGS. **4** and **7**, in this embodiment, the unlocking assembly **43** further includes a second handle **435**. The second handle **435** is disposed at one end of the pull rod body **432** facing away from the first mounting frame **431**. The second handle **435** can be used to facilitate the operators to grasp the pull rod body **432** and push the first mounting frame **431** to move relative to the floating tray **42**.

(34) Referring to FIGS. **7** and **8**, in this embodiment, the first mounting frame **431** is disposed at one end of the pull rod body **432**. The first mounting frame **431** has a substantially symmetrical structure. The first mounting frame **431** includes two spaced support plates **431a**, a support wing **431b** disposed at one end of each support plate **431a** facing away from the pull rod body **432**, and a connecting plate **431d** connected between the two support plates **431a**. Each support wing **431b** is substantially vertically connected to the support plate **431a** and extends backward from the pull rod body **432**. Each support wing **431b** defines a shaft groove **431c** on a side facing away from the pull rod body **432**.

(35) Referring to FIG. **8**, the hook assembly **44** includes a second mounting frame **441**. The second mounting frame **441** is substantially inverted T-shaped. The second mounting frame **441** includes a crossbar **441a**, a column **441b** disposed on one side of the crossbar **441a**, and hooks **442** formed by substantially vertically extending from both ends of the crossbar **441a**. Opposite sides of the column **441b** are provided with first rotating shafts **443**. Each first rotating shaft **443** is rotatably accommodated in the corresponding shaft groove **431c**.

(36) Referring to FIGS. **8** and **9**, the hook **442** is substantially L-shaped. The hook **442** includes a first portion **442a**, a second portion **442b**, and a third portion **442c**. The second portion **442b** is connected between the first portion **442a** and the third portion **442c**. The third portion **442c** extends substantially perpendicular to the second portion **442b** and is used to hook the protrusion **312**. The first portion **442a** is connected to the crossbar **441a**, and one side of the first portion **442a** extending away from the other hook **442** is provided with a second rotating shaft **444**. The hook **442** is rotatably connected to the first side panel **422** of the floating tray **42** through the second rotating shaft **444**. When the unlocking assembly **43** is pushed, the support wing **431b** can push the column **441b** to cause the first portion **442a** of the hook **442** to rotate around the first side panel **422**. The rotation of the first portion **442a** can drive the third portion **442c** to rotate, and the third portion **442c** can hook the protrusion **312** of the unlocking slider **31** (FIG. **10**).

(37) Referring to FIG. **11**, in this embodiment, the hook assembly **44** further includes a third elastic member **445**. The third elastic member **445** is connected between the second portion **442b** and the first side panel **422** of the floating tray **42**, so that when the hook **442** rotates around the second rotating shaft **444**, the third elastic member **445** can obtain elastic potential energy to facilitate the hook **442** to return to a position before rotation.

(38) The embodiments shown and described above are only examples. Many details are often found in the art such as the other features of the device **40**. Therefore, many such details are neither shown nor described. Even though numerous characteristics and advantages of the present technology have been set forth in the foregoing description, together with details of the structure and function of the present disclosure, the disclosure is illustrative only, and changes may be made in the detail, especially in matters of shape, size and arrangement of the parts within the principles of the present disclosure up to, and including the full extent established by the broad general meaning of the terms used in the claims. It will therefore be appreciated that the embodiments described above may be modified within the scope of the claims.

Claims

1. An operation and maintenance device applied in a chassis, a storage tray is connected to the chassis through a slidable locking mechanism, the operation and maintenance device comprising: a floating tray; an unlocking assembly movably disposed on the floating tray, the unlocking assembly comprising a first mounting frame; a hook assembly connected to the unlocking assembly, the hook assembly comprising a second mounting frame and a hook connected to the second mounting frame, wherein the second mounting frame is rotatably connected to the first mounting frame, and is configured to drive the hook to rotate, thereby causing the hook to be engaged with the slidable locking mechanism, the unlocking assembly is configured to drive the hook to move, thereby causing the slidable locking mechanism to disengage a connection between the chassis and the storage tray; and a base, wherein the base comprises a first bottom plate, two first side plates, a second side plate, and a plurality of positioning pins, wherein the two first side plates are disposed on opposite sides of the first bottom plate, the second side plate are connected between the two first side plates; and the first bottom plate, the two first side plates, and the second side plate together form a first accommodating space, the floating tray is movably disposed in the first accommodating space, and the plurality of the positioning pins is disposed at ends of the two first side plates away from the second side plate.
2. The operation and maintenance device of claim 1, wherein the floating tray comprises a second bottom plate, two first side panels, a second side panel, and a handle, the two first side panels are disposed on opposite sides of the second bottom plate, the second side panel is connected between the two first side panels; the second bottom plate, the two first side plates, and the second side plate together form a second accommodating space; and the unlocking assembly is movably disposed in the second accommodating space, and the handle is disposed on a side of the second side plate away from the second accommodating space.
3. The operation and maintenance device of claim 2, wherein the second side plate defines an opening, and the handle extends through the opening.
4. The operation and maintenance device of claim 2, wherein a side of one of the first side panels is provided with a support column away from the second accommodating space, a side of another one of the first side plates is provided with a support bar facing the first accommodating space, the support column is movably disposed on the support bar.
5. The operation and maintenance device of claim 3, wherein the base further comprises a first fixing pin and a second fixing pin, the first fixing pin and the second fixing pin are disposed on a side of the first bottom plate facing the first accommodating space, and the first fixing pin and the second fixing pin are spaced from each other; and the floating tray further comprises a first elastic member and a third fixing pin, the third fixing pin is disposed on a side of the second bottom plate facing the second accommodating space, the second bottom plate defines a first sliding slot and a second sliding slot, the first fixing pin movably extends through the first sliding slot, the second fixing pin movably extends through the second sliding slot, and the first elastic member is connected between the first fixing pin and the third fixing pin.
6. The operation and maintenance device of claim 4, wherein the floating tray further comprises a first fixing column and a second fixing column, the first fixing column and the second fixing column are disposed on a side of the second bottom plate facing the second accommodating space, and the first fixing column and the second fixing column are spaced from each other; and the unlocking assembly further comprises a pull rod body, a second elastic member, and a third fixing column, the third fixing column is disposed on a side of the pull rod body away from the second accommodating space, the pull rod body defines a first limiting slot and a second limiting slot, the first fixing column movably extends through the first limiting slot, the second fixing column movably extends through the second limiting slot, and the second elastic member is connected

between the first fixing column and the third fixing column.

7. The operation and maintenance device of claim 1, wherein the first mounting frame comprises a support plate and a support wing connected to the support plate, one end of the support plate away from the support wing is connected to the unlocking assembly, and the support wing defines a shaft groove; the hook assembly further comprises a first rotating shaft and a second rotating shaft, the first rotating shaft is disposed at one end of the second mounting frame away from the hook, and the first rotating shaft is rotatably accommodated in the shaft groove; and the hook comprises a first portion, a second portion, and a third portion, the first portion is connected to another end of the second mounting frame, the second portion is connected between the first portion and the third portion, and the first portion is rotatably connected to one end of the floating tray through the second rotating shaft.

8. The operation and maintenance device of claim 7, wherein the hook assembly further comprises a third elastic member, and the third elastic member is connected between the second portion and the floating tray.

9. The operation and maintenance device of claim 7, wherein the second mounting frame comprises a crossbar and a column disposed on one side of the crossbar, one end of the crossbar is connected to the hook, one side of the column is connected to the first rotating shaft.

10. A cabinet comprising: a chassis; a storage tray removably disposed in the chassis; a slidable locking mechanism connecting the chassis and the storage tray; and an operation and maintenance device comprising: a floating tray; an unlocking assembly movably disposed on the floating tray, the unlocking assembly comprising a first mounting frame; and a hook assembly connected to the unlocking assembly, the hook assembly comprising a second mounting frame and a hook connected to the second mounting frame, wherein the second mounting frame is rotatably connected to the first mounting frame, and is configured to drive the hook to rotate, thereby switching the hook between a first state and a second state; wherein the first state is that the hook is disengaged from the slidable locking mechanism, and the slidable locking mechanism connects the storage tray to the chassis; the second state is that the hook is engaged with the slidable locking mechanism, and the slidable locking mechanism unlocks the storage tray and the chassis; the operation and maintenance device further comprising a base, wherein the base comprises a first bottom plate, two first side plates, a second side plate, and a plurality of positioning pins; the two first side plates are disposed on opposite sides of the first bottom plate, the second side plate are connected between the two first side plates; and the first bottom plate, the two first side plates, and the second side plate together form a first accommodating space, the floating tray is movably disposed in the first accommodating space, and the plurality of the positioning pins is disposed at ends of the two first side plates away from the second side plate.

11. The cabinet of claim 10, wherein the slidable locking mechanism comprises: an elastic pin disposed on the storage tray, the elastic pin comprises a first end and a second end, the first end extends through the storage tray to insert into a pin hole in the chassis, a cross-sectional dimension of the first end is less than a cross-sectional dimension of the second end; and an unlocking slider defines an slanted slot, the elastic pin movably extends through the slanted slot, the second end abuts against an inner wall of the slanted slot, and an extending direction of the slanted slot is inclined with respect to an extending direction of the elastic pin, a cross-sectional dimension of the slanted slot is greater than the cross-sectional dimension of the first end and less than the cross-sectional dimension of the second end.

12. The cabinet of claim 10, wherein the floating tray comprises a second bottom plate, two first side panels, a second side panel, and a handle, the two first side panels are disposed on opposite sides of the second bottom plate, the second side panel is connected between the two first side panels; the second bottom plate, the two first side plates, and the second side plate together form a second accommodating space; and the unlocking assembly is movably disposed in the second accommodating space, and the handle is disposed on a side of the second side plate away from the

second accommodating space.

13. The cabinet of claim 12, wherein the second side plate defines an opening, and the handle extends through the opening.

14. The cabinet of claim 12, wherein a side of one of the first side panels is provided with a support column away from the second accommodating space, a side of another one of the first side plate is provided with a support bar facing the first accommodating space, the support column is movably disposed on the support bar.

15. The cabinet of claim 13, wherein the base further comprises a first fixing pin and a second fixing pin, the first fixing pin and the second fixing pin are disposed on a side of the first bottom plate facing the first accommodating space, and the first fixing pin and the second fixing pin are spaced from each other; and the floating tray further comprises a first elastic member and a third fixing pin, the third fixing pin is disposed on a side of the second bottom plate facing the second accommodating space, the second bottom plate defines a first sliding slot and a second sliding slot, the first fixing pin movably extends through the first sliding slot, the second fixing pin movably extends through the second sliding slot, and the first elastic member is connected between the first fixing pin and the third fixing pin.

16. The cabinet of claim 10, wherein the floating tray further comprises a first fixing column and a second fixing column, the first fixing column and the second fixing column are disposed on a side of the second bottom plate facing the second accommodating space, and the first fixing column and the second fixing column are spaced from each other; and the unlocking assembly further comprises a pull rod body, a second elastic member, and a third fixing column, the third fixing column is disposed on a side of the pull rod body away from the second accommodating space, the pull rod body defines a first limiting slot and a second limiting slot, the first fixing column movably extends through the first limiting slot, the second fixing column movably extends through the second limiting slot, and the second elastic member is connected between the first fixing column and the third fixing column.

17. The cabinet of claim 16, wherein the first mounting frame comprises a support plate and a support wing connected to the support plate, one end of the support plate away from the support wing is connected to the unlocking assembly, and the support wing defines a shaft groove; the hook assembly further comprises a first rotating shaft and a second rotating shaft, the first rotating shaft is disposed at one end of the second mounting frame away from the hook, and the first rotating shaft is rotatably accommodated in the shaft groove; and the hook comprises a first portion, a second portion, and a third portion, the first portion is connected to another end of the second mounting frame, the second portion is connected between the first portion and the third portion, and the first portion is rotatably connected to one end of the floating tray through the second rotating shaft.

18. The cabinet of claim 16, wherein the hook assembly further comprises a third elastic member, and the third elastic member is connected between the second portion and the floating tray.
